

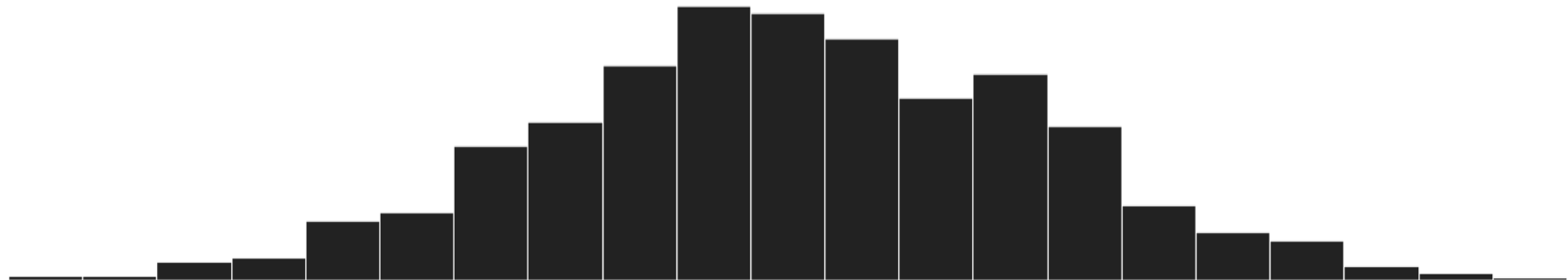
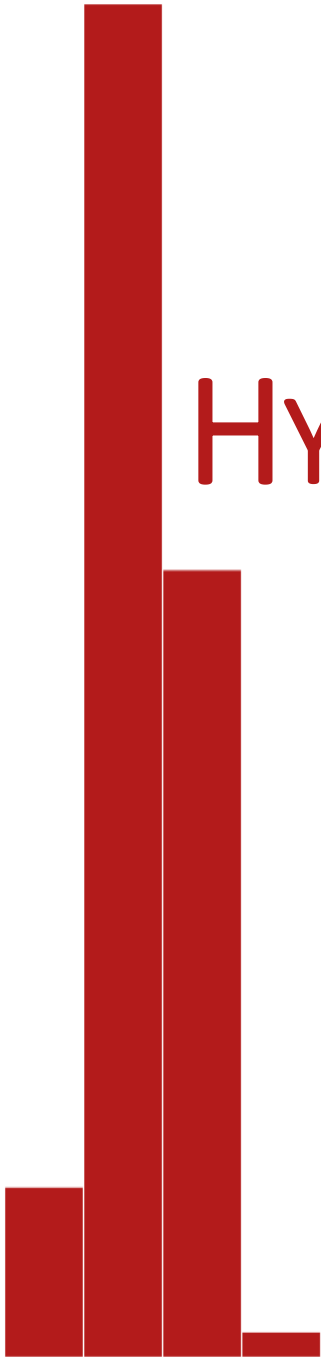
VTPEH 6105

Lecture 10

HYPOTHESIS TESTING & PREDICTIONS: LOGISTIC REGRESSIONS

Amandine Gamble

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House Keeping

Final Exam

- Write the review of a scientific publication with a focus on data analysis and interpretation. Limited guiding questions: you should be able to identify the strengths and limitations of the publication, and assess whether you trust the conclusion(s)
- No access to AI, Google or other online resources
- Paper, hand-written cheat sheet (any shape, color, language...)

Will be checked by the teaching team

Suggestions:

- Which test for what
- Assumptions for each test
- Best practice for data reporting and analysis
- Homework guiding questions

More About the Peer-Review Part of the Homework

- Some good and some bad publications
- You should now have tools to identify the good from the bad ones

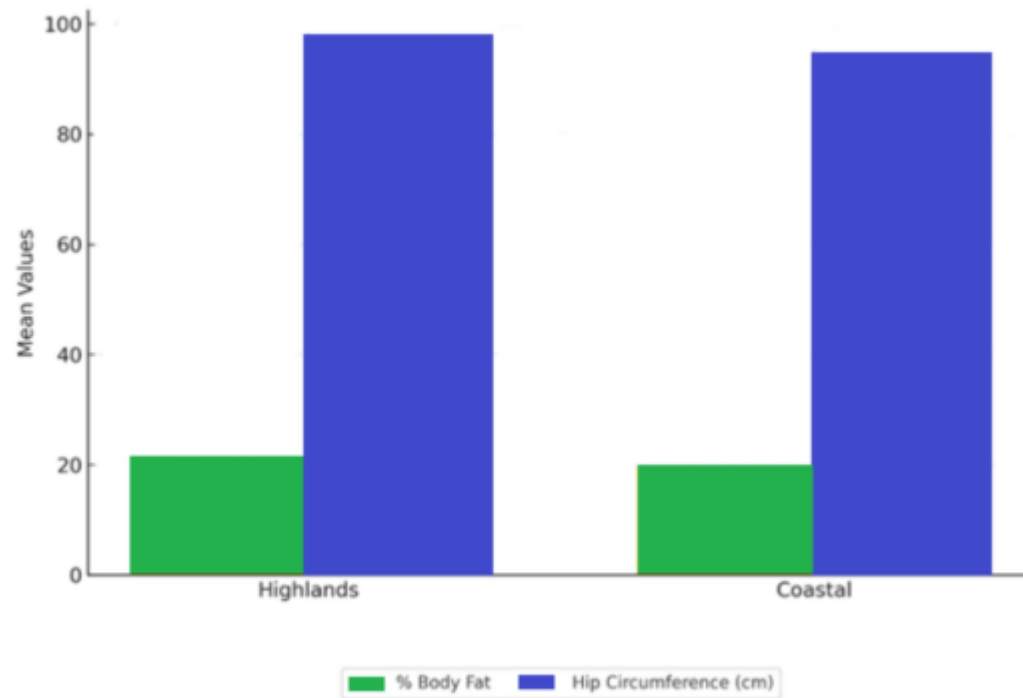
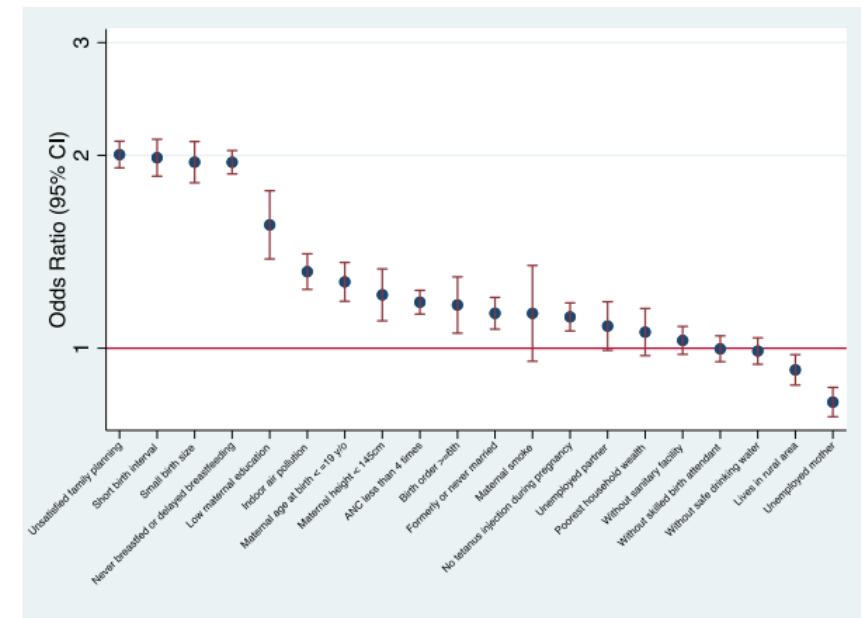


FIGURE 1
Body composition by birthplace.



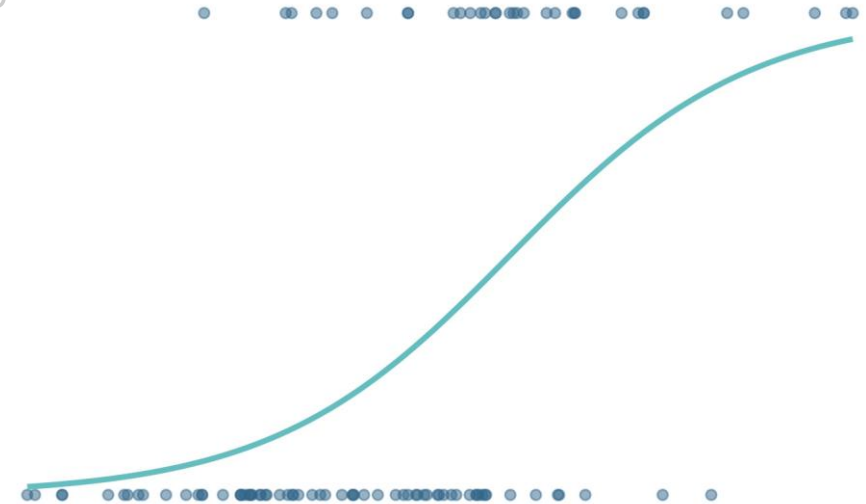
* Child sex, child age, and country-fixed effects were adjusted in this model
* Short birth interval – birth interval <18m; Low maternal education – mothers with no schooling;
ANC less than 4 times – antenatal care visit less than 4 times; without IFA supplementary during pregnancy – without iron/folic acid supplementary during pregnancy
*95% CI: 95% confidence interval

Fig. 1: Relative ranking of risk factors associated with under-5 mortality from mutually-adjusted models, pooled analysis of 61 countries (N = 604,372).

Logistic Regressions

Today's themes

- Moving from linear regressions to logistic regressions...
 - What are logistic regressions used for?
 - How do they work?
 - What are the assumptions?
 - How do we interpret the generated coefficients?



Linear regression vs logistic regression

Linear regression

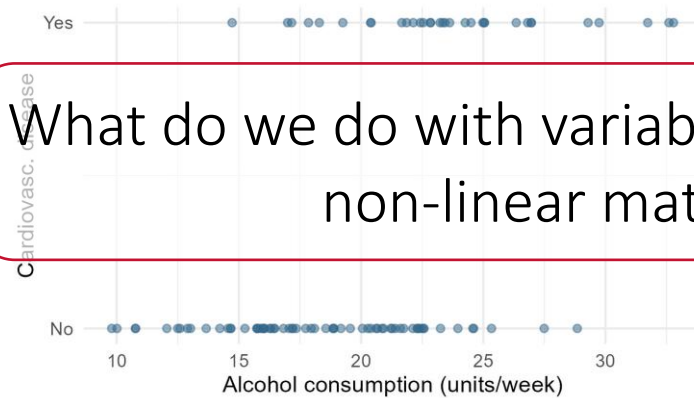
- The response variable Y is a quantitative variable
- The explanatory variables can be quantitative or categorical

Logistic regression

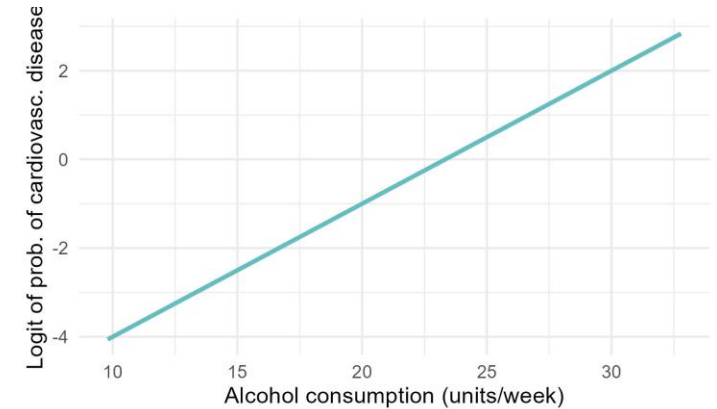
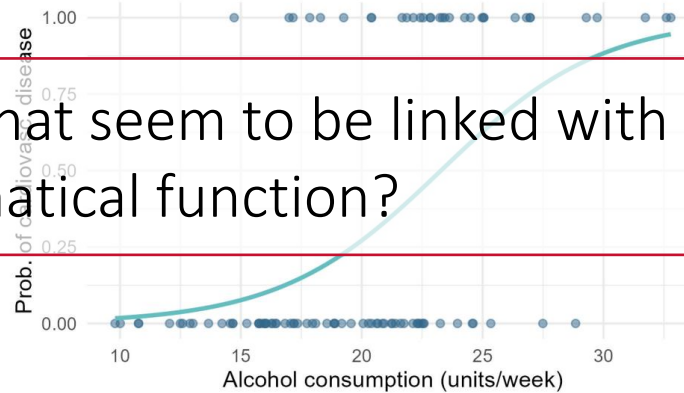
- The response variable Y is a **binary** variable
- The explanatory variables can be quantitative or categorical

Linear regression vs logistic regression

- Illustration: let's look at the relationship between the occurrence of cardiovascular diseases and alcohol consumption



What do we do with variables that seem to be linked with a non-linear mathematical function?



Occurrence of
cardiovascular diseases
 $Y \sim \text{Binomial}(p)$

Probability of
cardiovascular disease
 $p \sim \text{Sigmoid}$

Logit of the probability of
cardiovascular disease
$$\ln \left(\frac{p}{1-p} \right) \sim \alpha + \beta X$$

Linear regression vs logistic regression

Linear regression

- The response variable Y is a quantitative variable
- The explanatory variables can be quantitative or categorical
- The relationship between the response and explanatory variables can be written as

$$Y = \alpha + \beta_k X_k$$

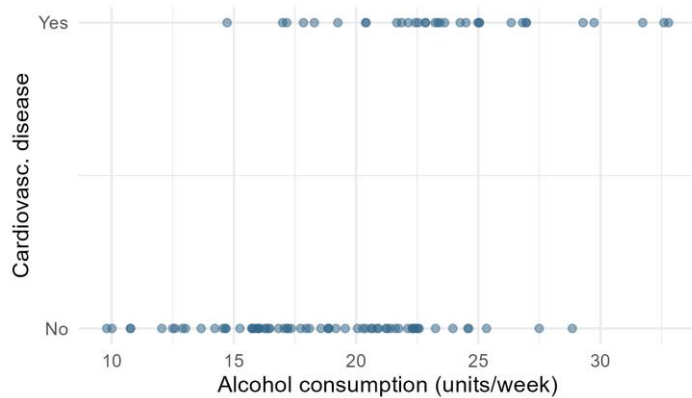
Logistic regression

- The response variable Y is a **binary** variable
- The explanatory variables can be quantitative or categorical
- With **p the probability of $Y = 1$** , the relationship between the response and explanatory variables can be written as

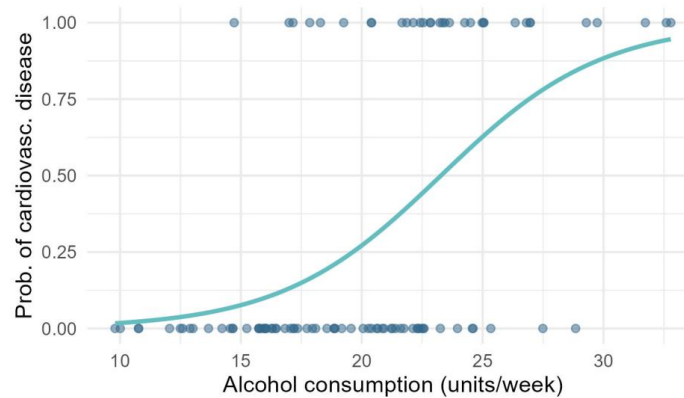
$$\ln \left(\frac{p}{1-p} \right) = \alpha + \beta_k X_k$$

Linear regression vs logistic regression

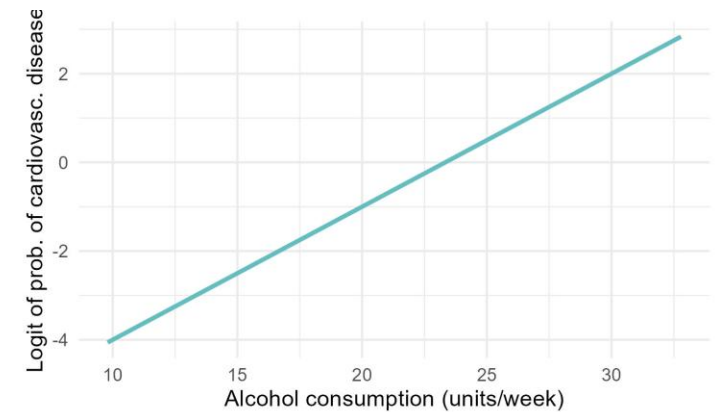
A logistic regression is a linear regression on a **logit-transformed response variable**, which is part of the broader category of **generalized linear models**



Occurrence of
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 $Y \sim \text{Binomial}(p)$



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Linear regression vs logistic regression

Linear regression

- The response variable Y is a quantitative variable
- The explanatory variables can be quantitative or categorical
- The relationship between the response and explanatory variables can be written as

$$Y = \alpha + \beta_k X_k$$

- The null hypothesis is H_0 : all regression coefficients (β) are equal to zero

Logistic regression

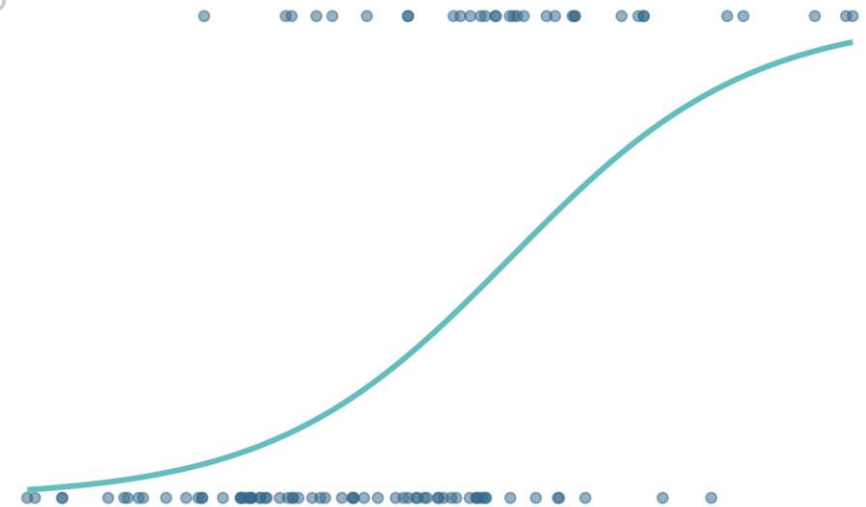
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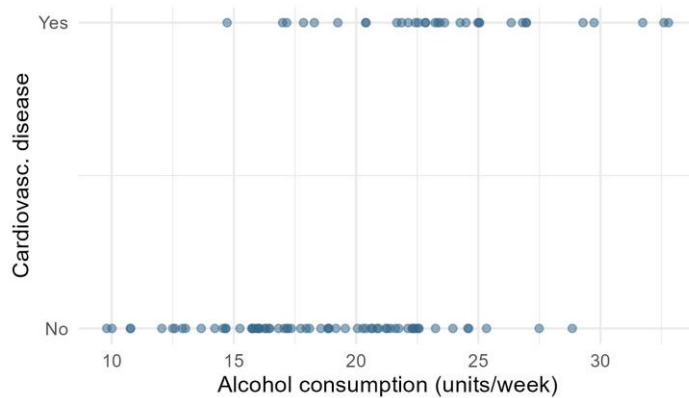
Today's themes

- Moving from linear regressions to logistic regressions...
 - What are logistic regressions used for?
 - How do they work?
 - What are the assumptions?
 - How do we interpret the generated coefficients?

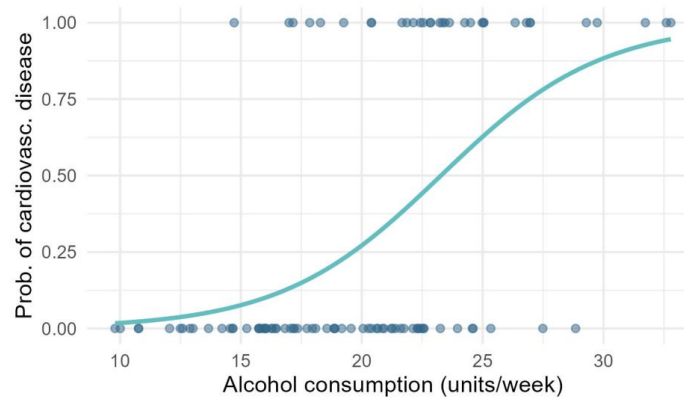


Linear regression vs logistic regression

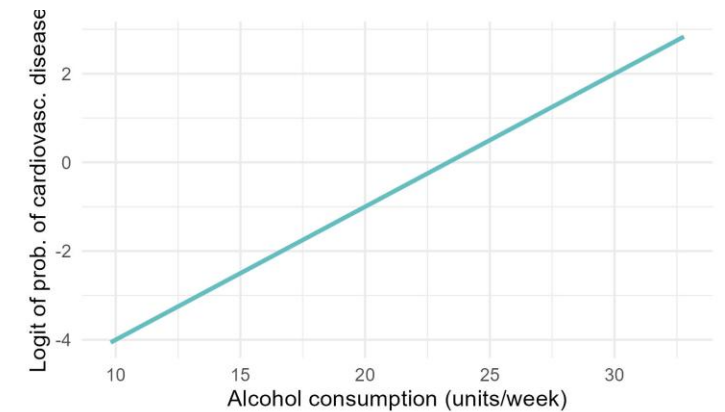
A logistic regression is a linear regression on a **logit-transformed variable**, which is part of the broader category of **generalized linear models**



Occurrence of
cardiovascular diseases
 $Y \sim \text{Binomial}(p)$



Probability of
cardiovascular disease
 $p \sim \text{Sigmoid}$



Logit of the probability of
cardiovascular disease

Odds of
cardiovascular
disease

$$\ln\left(\frac{p}{1-p}\right) = \alpha + \beta X$$

Reminder about odds...

	Case Yes for cardiovascular diseases	Control No for cardiovascular diseases
Exposed	a	b
Not exposed	c	d

- Odds of cardiovascular disease = $\frac{a+c}{b+d} = \frac{p}{1-p}$
With p the probability of cardiovascular disease

Logistic regressions: how?

A logistic regression is a linear regression on a **logit-transformed response variable**, which is part of the broader category of **generalized linear models**

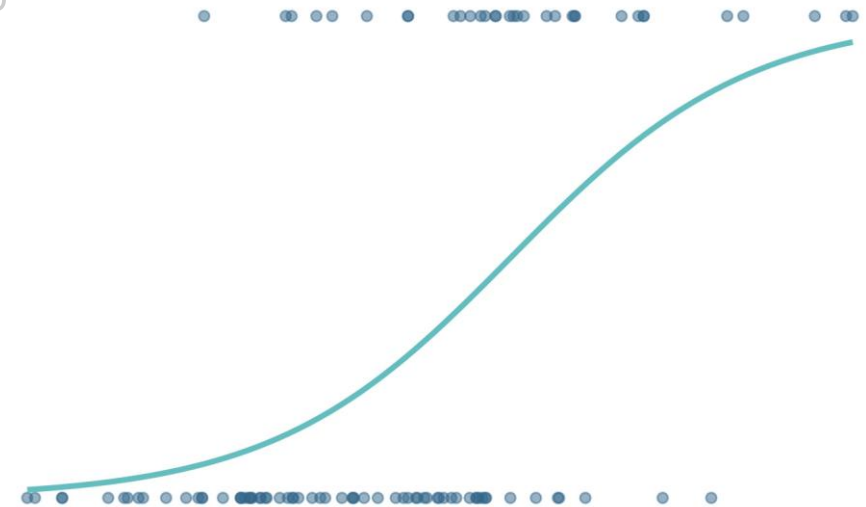
- A linear regression method is applied to the logit-transformed response variable...
- ... But not the ordinary least square as there are no errors (or residuals) *per se*
- Instead: maximum likelihood estimation (MLE)

$$\text{If } \ln\left(\frac{p}{1-p}\right) = \alpha + \beta_k X_k, \text{ then } p = \frac{e^{\alpha + \beta_k X_k}}{1 + e^{\alpha + \beta_k X_k}}$$

MLE finds the α and β_k that best explain the data (*i.e.*, that are the most likely considering the data)

Today's themes

- Moving from linear regressions to logistic regressions...
 - What are logistic regressions used for?
 - How do they work?
 - What are the assumptions?
 - How do we interpret the generated coefficients?



Logistic regressions: assumptions

- **Independence:** Y values are statistically independent (via ε_i)
- **Linearity between X s and $\text{logit}(\frac{P(Y=1)}{P(Y=0)})$:** the means of the subpopulations of Y all lie on the same straight line defined by
- ~~**Normality:** for each combinations of X s there is a subpopulation of Y values; these subpopulations must be normally distributed~~
- ~~**Homoscedasticity:** the variances of the subpopulations of Y are all equal and denoted by σ^2~~
- **Absence of co-linearity:** between covariates

Study design

Step 0 –
**Biological
assumption**

Step 1 – Visual
exploration of
the **raw data**

~~Step 2 – Visual
exploration of
the model
residuals~~

Logistic regressions: model selection

- As for linear models...

Trade-off between explained variance (the more covariates, the better) and biological relevance and meaningfulness (the least covariates the better)

Statistical conclusions can only be made after model selection as model structure impacts effect size estimates (and p-values)

The best model is the **most parsimonious**

Logistic regressions: interpretation

```
glm(formula = disease ~ alcohol, family = binomial, data = data)
```

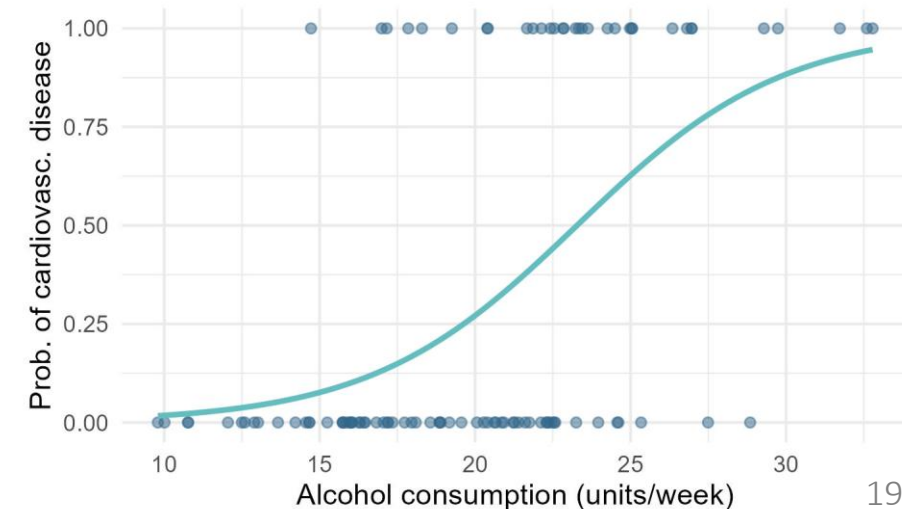
Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-7.01627	1.50052	-4.676	2.93e-06	***
alcohol	0.30142	0.06892	4.374	1.22e-05	***

- Assess associations: remember the null hypothesis...

H_0 : all regression coefficients (β) are equal to zero

- Make sense of the effect size: remember the transformation...



Logistic regressions: interpretation

- From regression coefficients to biologically meaningful numbers...

Let's simplify and consider $X = 0$ (does not drink alcohol) and $X = 1$ (drinks alcohol)

We defined $P(Y = 1|X) = p = \frac{e^{\alpha+\beta X}}{1+e^{\alpha+\beta X}}$, $P(Y = 0|X) = 1 - p = 1 - \frac{e^{\alpha+\beta X}}{1+e^{\alpha+\beta X}} = \frac{1}{1+e^{\alpha+\beta X}}$

$$\text{So } \frac{P(Y=1|X=1)/P(Y=0|X=1)}{P(Y=1|X=0)/P(Y=0|X=0)} = \frac{\frac{e^{\alpha+\beta \times 1}}{1+e^{\alpha+\beta \times 1}} / \frac{1}{1+e^{\alpha+\beta \times 1}}}{\frac{e^{\alpha+\beta \times 0}}{1+e^{\alpha+\beta \times 0}} / \frac{1}{1+e^{\alpha+\beta \times 0}}} = \frac{\frac{e^{\alpha+\beta}}{1+e^{\alpha+\beta}} / \frac{1}{1+e^{\alpha+\beta}}}{\frac{e^{\alpha}}{1+e^{\alpha}} / \frac{1}{1+e^{\alpha}}} = \frac{e^{\alpha+\beta}}{e^{\alpha}} = e^{\alpha+\beta-\alpha} = e^{\beta}$$

Odds ratio

The exponential of the coefficient corresponds to the **odds ratio**

Reminder about odds...

	Case Yes for cardiovascular diseases	Control No for cardiovascular diseases
Exposed	a	b
Not exposed	c	d

- Odds of cardiovascular disease = $\frac{a+c}{b+d} = \frac{1}{1-p}$
With p the probability of cardiovascular disease
- Odds ratio of cardiovascular disease and alcohol consumption =
$$\frac{a/b}{c/d} = \frac{ad}{bc} = e^{\beta}$$

Logistic regressions: interpretation

- From regression coefficients to biologically meaningful numbers...
Let's generalize and consider $X = x$ and $x + 1$ weekly units of alcohol

$$\text{So } \frac{P(Y=1 | X=x) / P(Y=0 | X=x)}{P(Y=1 | X=x+1) / P(Y=0 | X=x+1)} = \frac{\frac{e^{\alpha+\beta}}{1+e^{\alpha+\beta}} / \frac{1}{1+e^{\alpha+\beta}}}{\frac{e^{\alpha+\beta+\beta x}}{1+e^{\alpha+\beta+\beta x}} / \frac{1}{e^{\alpha+\beta+\beta x}}} = \frac{e^{\alpha+\beta}}{e^{\alpha}} = e^{\alpha+\beta-\alpha} = e^{\beta}$$

Odds ratio

The exponential of the coefficient corresponds to the **odds ratio** for a difference of 1 unit of the explanatory variable
For every increase of 1 unit for the explanatory variable, the odds of the outcome is **multiplied by e^{β} with β the effect size**

Logistic regressions: interpretation

```
glm(formula = disease ~ alcohol, family = binomial, data = data)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-7.01627	1.50052	-4.676	2.93e-06	***
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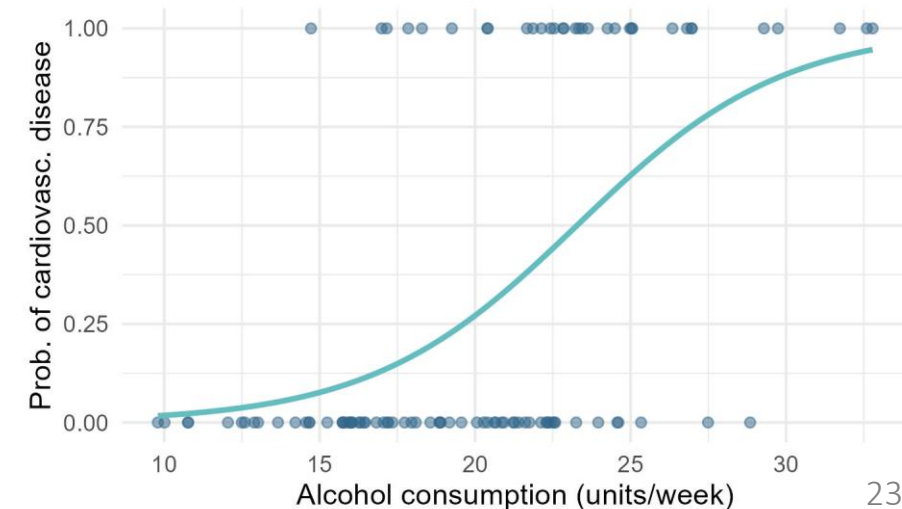
OR[95%CI] = 1.35[1.18; 1.55] > 1

- Assess associations: remember the null hypothesis...

H_0 : all regression coefficients (β) are equal to zero

- Make sense of the effect size: remember the transformation...

$$\text{OR}[95\%CI] = e^{\beta} [e^{\beta-1.96se}; e^{\beta+1.96se}]$$



Logistic regressions: conclusion

- As for linear models...

Check the effect sizes (correlations are meaningful?)

The exponential of the coefficient corresponds to the **odds ratio** for a difference of 1 unit of the explanatory variable
For every increase of 1 unit for the explanatory variable, the odds of the outcome is **multiplied by e^{β} with β the effect size**

- A coefficient $\neq 0$, or OR $\neq 1$, indicates a significant association
 - $\beta > 0$, or OR > 1 : positive association
 - $\beta < 0$, or OR < 1 : negative association

Logistic regressions: predictions

- We know the OR, which allows us to assess associations between the explanatory and the response variables

$$\mathbf{OR}_k = e^{\beta_k}$$

- How to make predictions?
- We also know $\mathbf{p} = \mathbf{P}(Y = \mathbf{1} \mid \mathbf{X}_k)$, the probability of “success” (or “occurrence”) as a function of the explanatory variables ($\mathbf{X}_k$)

$$\ln\left(\frac{p}{1-p}\right) = \alpha + \beta_k X_k \Leftrightarrow \mathbf{p} = \frac{e^{\alpha + \beta_k X_k}}{1 + e^{\alpha + \beta_k X_k}}$$

We cannot predict the outcome (Y), but we can predict the probability of occurrence as a function of the explanatory variables ($\mathbf{X}_k$)

$$p = P(Y = 1) = \frac{e^{\alpha + \beta_k X_k}}{1 + e^{\alpha + \beta_k X_k}}$$

Logistic regressions: predictions

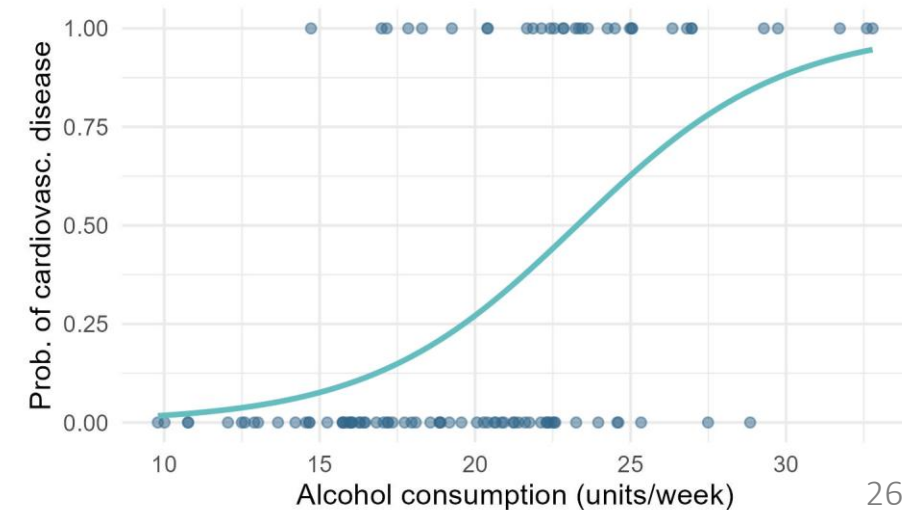
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Coefficients:

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alcohol	0.30142	0.06892	4.374	1.22e-05	***

OR[95%CI] = 1.35[1.18; 1.55] > 1

$$p(\text{Disease} \mid \text{Alcohol}) = \frac{e^{-7.02+0.3 \times \text{Alcohol}}}{1 + e^{-7.02+0.3 \times \text{Alcohol}}}$$



Logistic regressions: predictions

```
glm(formula = disease ~ alcohol, family = binomial, data = data)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-7.01627	1.50052	-4.676	2.93e-06	***
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OR[95%CI] = 1.35[1.18; 1.55] > 1

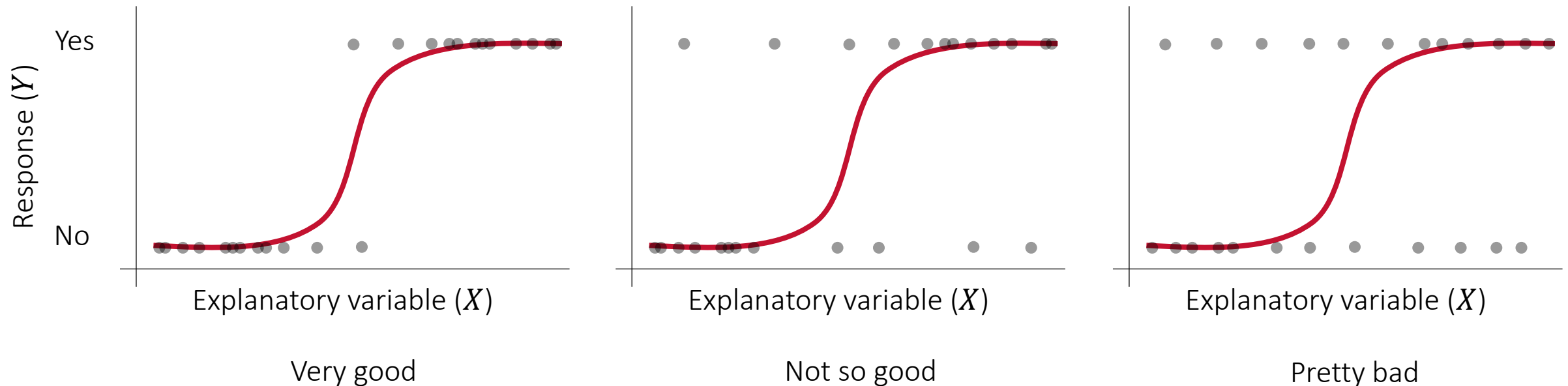
- With uncertainty:

95%CI of $p(\mathbf{Disease} \mid \mathbf{Alcohol}) =$

$$\left[\frac{e^{-7.02 - 1.96 \times 1.50 + (0.3 - 1.96 \times 0.07) \times \text{Alcohol}}}{1 + e^{-7.02 - 1.96 \times 1.50 + (0.3 - 1.96 \times 0.07) \times \text{Alcohol}}} ; \frac{e^{-7.02 + 1.96 \times 1.50 + (0.3 + 1.96 \times 0.07) \times \text{Alcohol}}}{1 + e^{-7.02 + 1.96 \times 1.50 + (0.3 + 1.96 \times 0.07) \times \text{Alcohol}}} \right]$$

Goodness of fit for logistic regression

- As usual, we can always run a model, even a bad model, so let's assess its quality through its "goodness of fit"



Logistic Regressions

Additional points...

Logistic regressions: when?

- When we want to investigate the relationship between a binary response variable and...
 - A quantitative explanatory variable
 - Several (quantitative or qualitative) variables
- Otherwise...
 - Chi^2
- If the response variable has more than 2 outcomes...
 - Multinomial logistics regression
 - Ordinal logistic regression

Logistic regressions: model selection

- As for linear models...

Trade-off between explained variance (the more covariates, the better) and biological relevance and meaningfulness (the least covariates the better)

Statistical conclusions can only be made after model selection as model structure impacts effect size estimates (and p-values)

The best model is the **most parsimonious**

Multivariate logistic regression

- Illustration: let's look at the relationship between the occurrence of cardiovascular diseases and several potential predictors
 - Alcohol consumptions (units/week)
 - Gender (Woman or Man)
 - Stress level (score)
 - Instagram followers (number)

Logistic regressions: assumptions

- Independence: Y values are statistically independent (via ε_i)
- Linearity between X s and **logit**($P(Y = 1)$)
- Absence of co-linearity: between covariates



Study design



Step 0 –

**Biological
assumption**

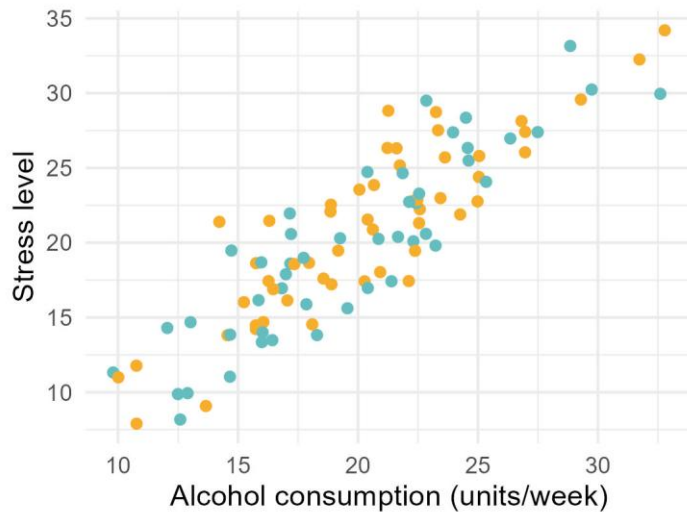
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Multivariate logistic regression

- Illustration: let's look at the relationship between the occurrence of cardiovascular diseases and several potential predictors

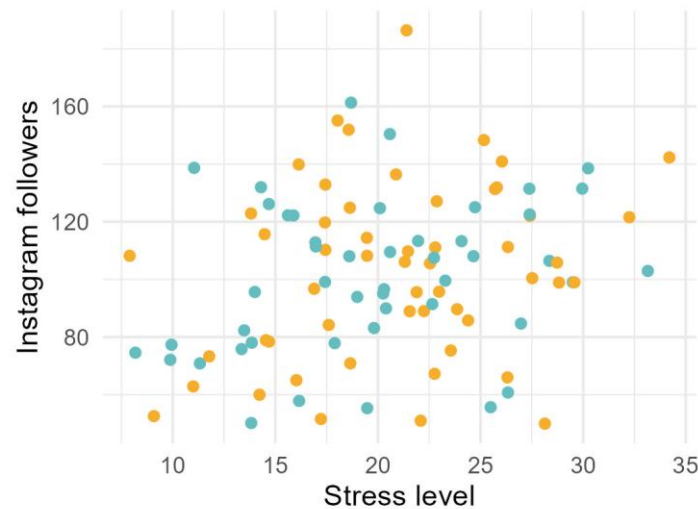
$\text{disease} \sim \text{alcohol} + \text{gender} + \text{stress} + \text{followers} ?$

- Testing assumptions... Collinearity



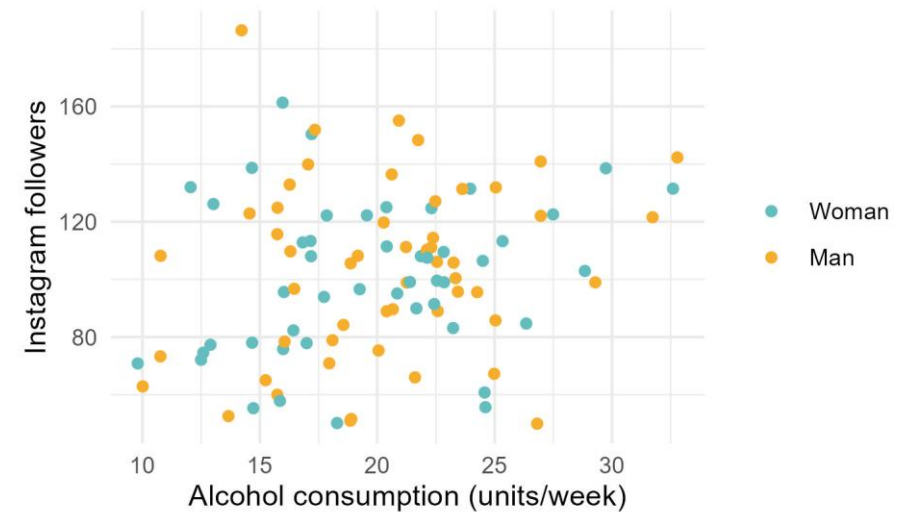
$$R^2 = 0.77$$

=/



$$R^2 = 0.04$$

=)

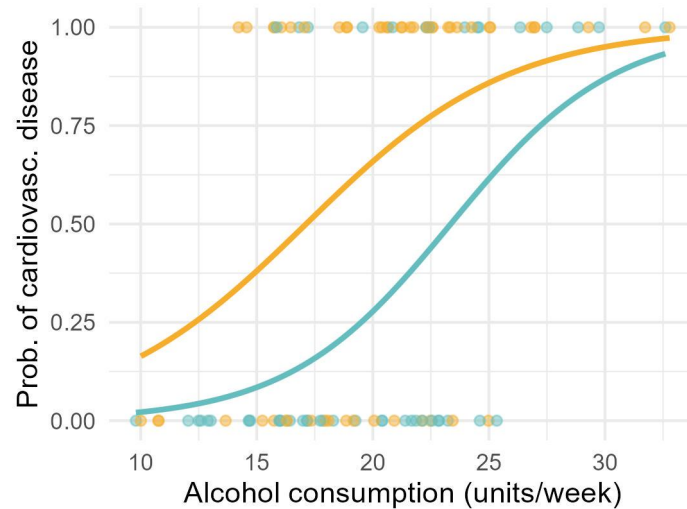


$$R^2 = 0.06$$

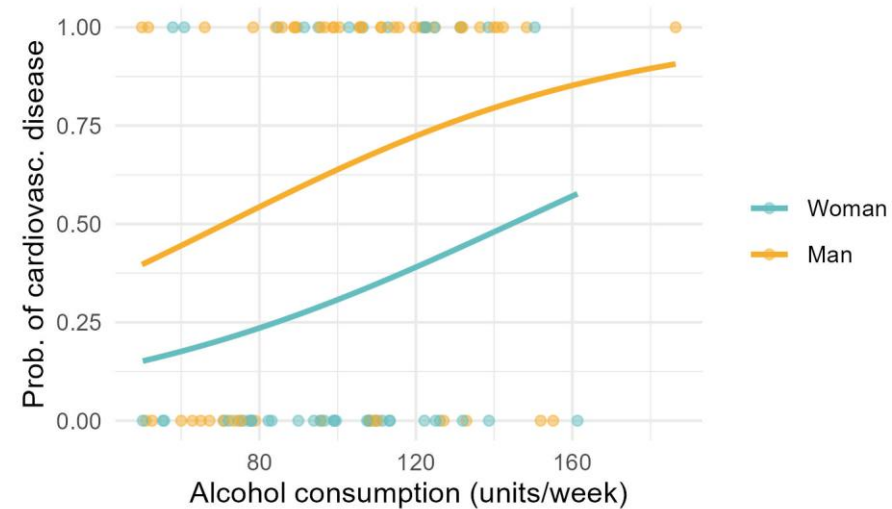
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Multivariate logistic regression

- Illustration: let's look at the relationship between the occurrence of cardiovascular diseases and several potential predictors
$$\text{disease} \sim \text{alcohol} + \text{gender} + \cancel{\text{stress}} + \text{followers} ?$$
- Testing assumptions... Linear relationship of the $\ln(\text{odds})$ (sigmoid of the raw data) with the explanatory variable



=)



?

Multivariate logistic regression

- Illustration: let's look at the relationship between the occurrence of cardiovascular diseases and several potential predictors

`disease ~ alcohol + gender` + ~~stress~~ + followers ?

- Model fitting and selection... Forward selection

```
model1 <- glm(disease ~ alcohol, data = data, family = binomial); summary(model1)
```

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-4.74862	1.15988	-4.094	4.24e-05	***
alcohol	0.23509	0.05689	4.132	3.60e-05	***

```
model2 <- glm(disease ~ alcohol + gender, data = data, family = binomial); summary(model2)
```

(Intercept)	-6.01468	1.35525	-4.438	9.08e-06	***
alcohol	0.25484	0.06156	4.140	3.47e-05	***
genderMan	1.59770	0.50045	3.193	0.00141	**

```
model3 <- glm(disease ~ alcohol + gender + stress, data = data, family = binomial); summary(model3)
```

(Intercept)	-6.01276	1.36536	-4.404	1.06e-05	***
alcohol	0.11042	0.10575	1.044	0.2964	
genderMan	1.56136	0.50758	3.076	0.0021	**
stress	0.14193	0.08855	1.603	0.1090	

```
model4 <- glm(disease ~ alcohol + gender + followers, data = data, family = binomial); summary(model4)
```

(Intercept)	-7.719363	1.760353	-4.385	1.16e-05	***
alcohol	0.255076	0.064647	3.946	7.96e-05	***
genderMan	1.618241	0.509323	3.177	0.00149	**
followers	0.016515	0.008928	1.850	0.06434	.

Multivariate logistic regression

- Illustration: let's look at the relationship between the occurrence of cardiovascular diseases and several potential predictors

`disease ~ alcohol + gender` + ~~stress~~ + followers ?

- Model fitting and selection... Forward selection

```
model2 <- glm(disease ~ alcohol + gender, data = data, family = binomial); summary(model2)
```

(Intercept)	-6.01468	1.35525	-4.438	9.08e-06	***
alcohol	0.25484	0.06156	4.140	3.47e-05	***
genderMan	1.59770	0.50045	3.193	0.00141	**

```
model2bis <- glm(disease ~ alcohol * gender, data = data, family = binomial); summary(model2)
```

(Intercept)	-6.65254	2.04028	-3.261	0.00111	**
alcohol	0.28483	0.09413	3.026	0.00248	**
genderMan	2.72689	2.59514	1.051	0.29337	
alcohol:genderMan	-0.05561	0.12475	-0.446	0.65576	

Multivariate logistic regression

- Illustration: let's look at the relationship between the occurrence of cardiovascular diseases and several potential predictors

`disease ~ alcohol + gender` + ~~stress~~ + followers ?

- Quality assessment

Hosmer and Lemeshow goodness of fit (GOF) test

data: data\$disease, fitted(model)

X-squared = 1.4747, df = 8, p-value = 0.9931

Multivariate logistic regression

- Illustration: let's look at the relationship between the occurrence of cardiovascular diseases and several potential predictors

`disease ~ alcohol + gender` + ~~stress~~ + followers ?

- Interpretation

```
model <- glm(disease ~ alcohol + gender, data = data, family = binomial); summary(model2)
```

(Intercept)	-6.01468	1.35525	-4.438	9.08e-06	***
alcohol	0.25484	0.06156	4.140	3.47e-05	***
genderMan	1.59770	0.50045	3.193	0.00141	**

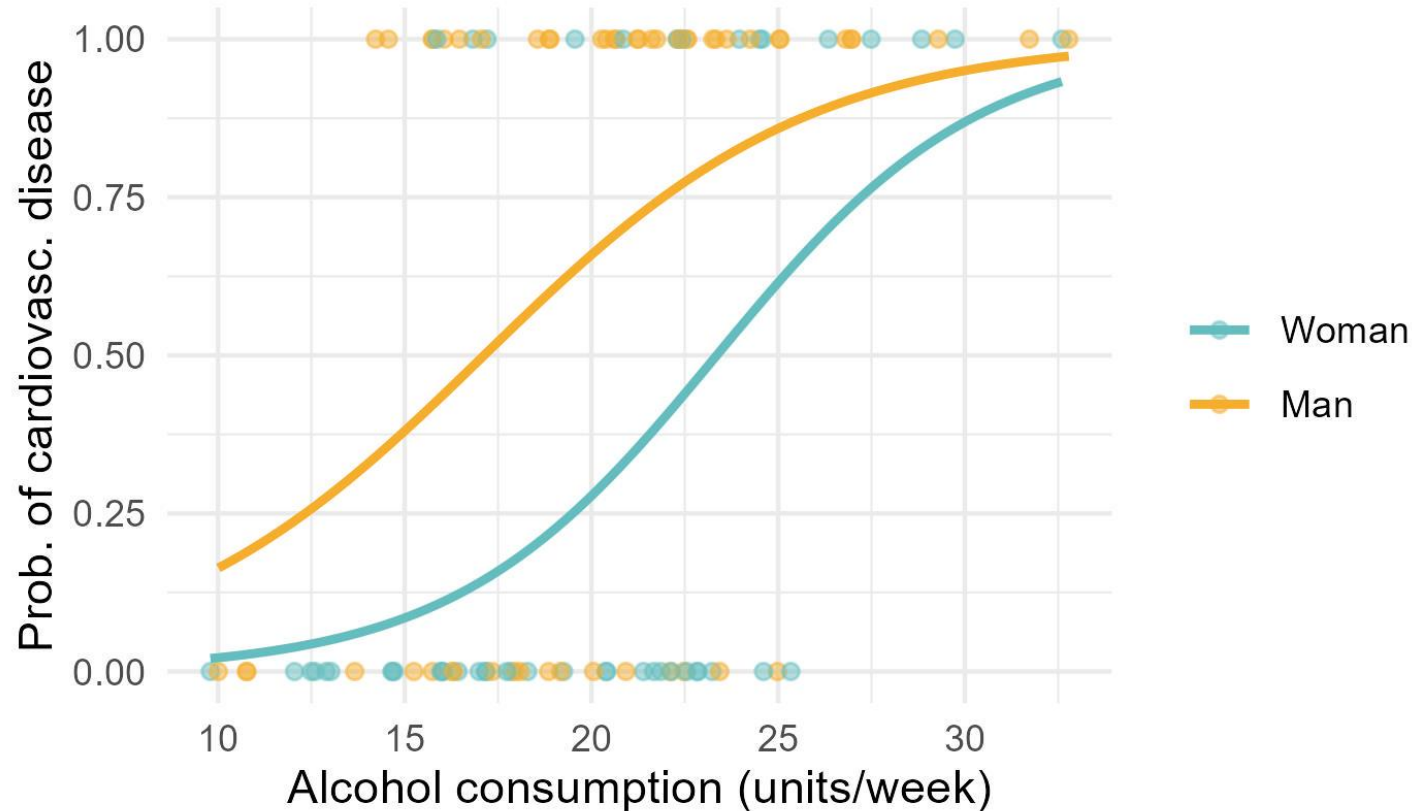
- Significant associations between cardiovascular disease occurrence and alcohol consumption and gender
- OR for alcohol: 1.29 [1.14 ; 1.46]
- OR for gender (Men compared to Women): 4.94 [1.85 ; 13.18]
- Predictions:

- For women: $p(\text{disease} \mid \text{alcohol}) = \frac{e^{-6 + 0.25 \times \text{alcohol}}}{1 + e^{-6 + 0.25 \times \text{alcohol}}}$
- For men: $p(\text{disease} \mid \text{alcohol}) = \frac{e^{-4.42 + 0.25 \times \text{alcohol}}}{1 + e^{-4.42 + 0.25 \times \text{alcohol}}}$

Multivariate logistic regression

- Illustration: let's look at the relationship between the occurrence of cardiovascular diseases and several potential predictors

$\text{disease} \sim \text{alcohol} + \text{gender} + \text{stress} + \text{followers} ?$



Before Next Time (Friday February 6, in LH3)

- Practice: **Homework #10** due after Spring Break
- Use the Guiding Quiz #10 if needed
- Check you are ready for next class: **Knowledge Check #10** due after Spring Break